

Physiological Stress Response of Hairy Shore Crab (*Hemigrapsus oregonensis*) to Acute Anthropogenic Noise Exposure

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Abstract

Noise pollution has been proven to be remarkably damaging to marine animals, preventing communication and interrupting key organismal processes. The Dungeness crab population, and by extension, industry, are at risk because of the invasive European green crab which is causing the Dungeness population to decrease. Knowing the damage noise pollution causes broadly, it is vitally important to minimize any anthropogenic impact on the Dungeness. A lack of current research and knowledge exists regarding invertebrates' response to noise pollution in its various forms, despite how harmful it is known to be for animals. Here we show that hairy shore crabs, closely related to Dungeness crabs, have no significant physiological stress response when exposed to acute anthropogenic noise pollution. It was previously understood that ship noises induced increased respiration rates in European green crabs, which was anticipated to be supported here. However, shore crabs reacted to acute noise exposure in an unexpected manner, not in line with the broader or invertebrate-specific conclusions already drawn. Our results demonstrate that not all anthropogenic noise is inherently harmful to invertebrates, and that factors such as noise type, duration, and frequency of exposure may play a highly important role. We anticipate that our findings will be used as evidence of a lack of knowledge and understanding surrounding invertebrate response to noise pollution. It cannot be concluded that invertebrates are not affected by noise pollution, given previous findings, and instead more specific information is required to guide conservation efforts and planning guidelines. Discovering vulnerability and stress response by invertebrates will aid in maintaining health of Dungeness crabs and saving the monetary value of the fishery.

1. Introduction

A valuable fishery is already showing signs of a struggle, and changing conditions are not in its favor. Dungeness Crab (*Metacarcinus magister*) catch within the state of Washington is extremely valuable, hauling in just more than 16 million pounds during the 2024-25 fishing season with an average price of \$6.85 per pound (WDFW, [2026](#)). This totals out to be over \$110 million of revenue. Shockingly enough, this massive sum was a 23% decline from the year prior, and 44% from two fishing seasons ago (WDFW, [2026](#)). The fishery is not at fault; the crabs themselves are under threat.

European Green Crabs (*Carcinus maenas*) are a dangerous invasive species that pose a threat to

shellfish and many other species (WDFW, [2025](#)).

When they have been allowed to establish a large population for an extended time in the past, they have dramatic impacts on other species, particularly smaller shore crabs (WDFW, [2025](#)). This means they put Dungeness crabs at high risk through competition for resources and direct consumption.

In an already vulnerable population, reducing other sources of damage is essential to protect this fishery. Anthropogenic noise pollution is a particularly concerning danger, alterable through direct human action. Evidence has already shown that anthropogenic noise negatively impacts marine animals (Duarte et al., [2021](#)). Sound serves as an essential sensory cue, and outside sources can “pollute” natural noise systems, damaging the

organisms that rely on it (Duarte et al., [2021](#)). Ship noise was concluded to increase respiration in European green crabs (Birch et al., [2025](#)). However, only a small amount of other research has been done to investigate the impact of noise pollution regarding invertebrates, a problem that needs to be investigated.

The hairy shore crab (*Hemigrapsus oregonensis*) is a small invertebrate, found to be a very frequent prey of the European green crab (Fisher et al., [2024](#)). They are also native to Washington state, with key similarities to the Dungeness crab (WDFW, [2023](#)). They possess 10 jointed limbs, abdomen flaps, and share the same body plan (WDFW, [2023](#)). Because of these similarities, we chose to approximate the physiological stress response to noise exposure of these organisms to that of a Dungeness crab. Furthermore, the Dungeness crab rely on hairy shore crabs for ecosystem balance and food consumption. Damage to hairy shore crabs can be applied indirectly to Dungeness crabs. Our experimentation would put these hairy shore crabs through acute, intense noise exposure to simulate an event near a waterfront causing such a disturbance.

Based on a synthesis of the previously stated knowledge, we hypothesized that hairy shore crabs would undergo physiological stress in response to acute anthropogenic noise exposure, in the form of increased respiration and greater use of their internal resources. Therefore, the objective of this study was to determine if acute anthropogenic noise exposure induced a stress response in hairy shore crabs. We had three metrics used to evaluate their physiological state and look for a stress response: respiration rate, righting time, and glucose levels within their blood. This information could be applied to conservation efforts and regulations attempting to protect the Dungeness crab fishery and prevent any further unnecessary damage.

2. Materials and Methods

2.1. Experimental design

To assess the impact of acute noise exposure on the physiological state of the crabs, two distinct groups were established, the experimental noise group and the control group. 12 total hairy shore crabs were selected, with 6 placed in both groups. The experimental noise group was comprised of 6 males, while the control group contained 2 females and 4 males.

2.2. Living conditions

For the duration of the two-week experiment period, crabs were kept in tanks controlled for temperature and salinity. Saltwater was maintained at 35ppt, with temperature regulated at 10°C for all subjects. Tanks were a simulated isolated environment; movement outside was not possible for experimental subjects. All tanks remained in a tray, including bubblers for optimal living conditions, with 16 similar tanks in close proximity as a part of the same tray. Feeding was limited. Crabs had enough space to move, but living space was minimal. Lids were required to prevent escape.

2.3. Noise exposure details

For approximately two hours each week, crab tanks were removed from trays for experimentation. The experimental group tank was placed in a different room for noise exposure. The control group tank was placed in a noise-isolated environment, ensuring that all physiological stress responses of the control group were not due to noise. Once control group was in a secure environment, noise exposure started. Songs were played in a top to bottom order from a playlist accessible on Apple Music – Techno Rave by Magic Records – with song order preserved each week of exposure. Music was kept at an average of 72 dB, played from a JBL Go 4 Portable Bluetooth Speaker. Speaker was placed to the side of the long end of the rectangular tank, with output facing inward toward the water. Noise exposure continued in this manner for an hour.

2.4. Resazurin

After noise exposure began, resazurin wells were prepared. Two small 6-well plates were required, with these wells in a 2x3 arrangement. Each well had diameters of approximately 1.5 inches. Wells were then filled with 8 mL of resazurin working solution. Once filled, plates were moved to be near experimental and control setups. Additionally, two miniature plastic cups were filled with 8mL of resazurin working solution. One cup was placed near the control crabs and one near the experimental crabs, acting as blanks for each trial. With approximately 5 minutes left of the initial hour of noise exposure, T0 measurements were taken from each well and blank. 200 microliter aliquots were pipetted from each well, placed in a 96-well plate with 8 rows and 12 columns. 3 total aliquots were taken for each well and blank, meaning the top 3 rows were reserved for experimental and control crabs, with rows 5-7 reserved for the blanks. Subsequent measurements for each well were placed in the row below their initial measurement. Gloves were required to avoid skin exposure to resazurin working solution. Once the first hour of noise exposure was complete, both control and experimental crabs were placed within the 6-well plates, one well for each crab. Depth of wells was enough for full submergence, and lids were required to prevent escape. The experimental group was still exposed to noise while in the resazurin solution, with the speaker placed directly next to the plate. After 30 minutes, the second round of pipetting occurred, and 30 more was the final round of pipetting. 96-well plates should then be placed aside for absorbance measurement. Crabs were then removed from the resazurin wells and rinsed in saltwater twice. For the first week, after saltwater rinses they were tagged using glue and small stickers with numbers on them to attach their weekly measurements to their numbers for tracking. Control crabs were marked with yellow numbered stickers, and blue was used for experimental crabs. When all crabs were tagged and moved back to their respective tanks, tanks were placed back in the tray and used solution was discarded.

2.5. Righting Time

When the resazurin portion of data collection was complete and rinsing had been done, both groups were tested for their righting time. Subjects were flipped onto their backs and timed for how long it took them to “right” themselves back to their upright position in seconds. Time was collected using the Clock app pre-installed on apple devices (iOS 26.5). This process was performed one at a time.

2.6. Hemolymph extraction and analysis

For the third week of noise exposure only, crabs had blood samples drawn to be used in a glucose assay. Approximately 15 microliters were extracted from each crab, or the nearest amount of blood to this volume possible with one draw. Glucose assay was performed using a Cayman Chemical Glucose Colorimetric Assay Kit. Preparation completed in [accordance with SDS](#) accessible through Cayman Chemical. Absorbance was measured with a spectrophotometer to determine glucose levels in the blood upon extraction.

2.7. Graph development and calculations

All graphs and calculations were completed using a combination of R-studio version 4.5.3 and Microsoft Excel version 16.0. R package ggplot used to generate resazurin figures. Tukey’s method used to identify outliers in righting time ($1.5 * IQR$ on both ends).

3. Results

3.1. Resazurin fluorescence

There was no significant difference in fluorescence of resazurin wells discovered between control and experimental crab groups (Figure 1). Over the hour of resazurin submergence, control crabs had slightly higher average resazurin fluorescence across all weeks for both t1 and t2 measurements. However, this difference was not significant.

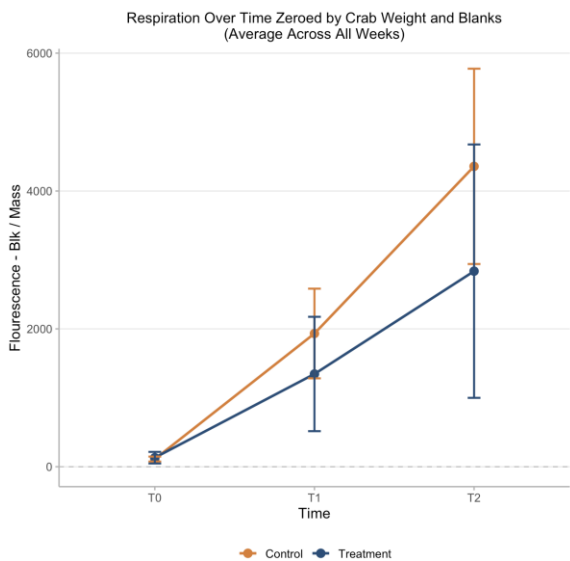


Figure 1: Fluorescence over hour of exposure for control and experimental groups of crabs. Experimental noise labeled as “treatment” in blue, control in yellow. Three different measurements of time on x-axis 30 minutes apart, fluorescence on y axis. Error bars for significance.

Looking at a week-to-week basis reveals similar findings (Figure 2). During week 1, t1 and t2 fluorescence was greater for the control group, but this difference was not significant. Week 2 had greater t1 and t2 fluorescence for the control group as well, with a significant difference from the experimental group. Week 3 had a slight difference in t0 measurements, the control group being slightly lower. t1 measurements were very similar, and the control group had a greater t2 difference, but this result was once again insignificant.

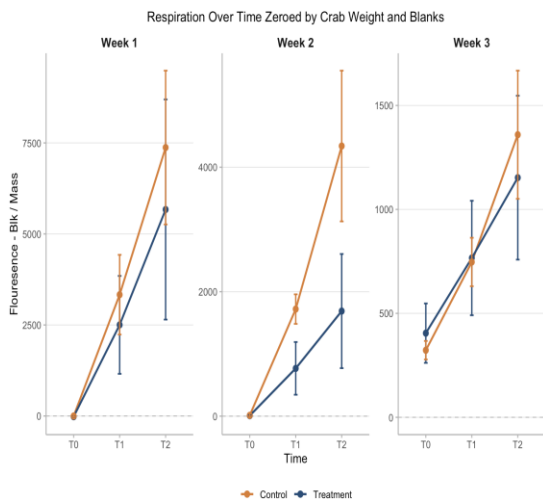


Figure 2: Fluorescence over time periods for all three weeks of exposure. Control in yellow, experimental labeled “treatment” in blue. Week labels found at top. Three time points for each week found at bottom.

3.2. Righting time

The righting time of the crabs within the experimental group was found to be higher than those within the control group (Figure 3). However, this difference was not significant. The control group had a median righting time of 0.53, and the experimental group had a median righting time of 0.935. Outliers were identified in the experimental group using Tukey’s method. An upper bound of 4.02 seconds was identified for the experimental group, and two observations were found above this level. These times were 46.81 seconds to right and 56.24 seconds to right. Although it is possible that the noise treatment is what caused these righting times, they were excluded from graphs because of the immense influence that they had on both statistical results and graph appearance. An upper bound of 2.55 seconds was identified for the control group. A single outlier was identified with this method, with a righting time of 4.69. However, without much swing and much closer to the upper bound, this was assumed reasonable and included in data.

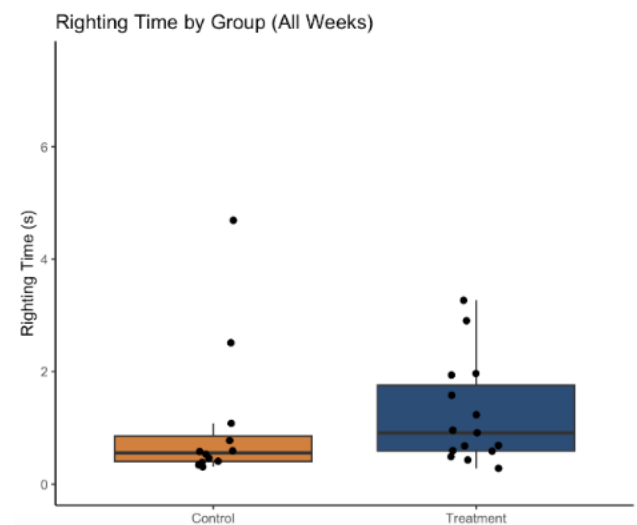


Figure 3: Righting time from all three weeks displayed by group. Righting time on y axis in seconds. X axis displays which group the box above depicts. Control in yellow, experimental in blue. Boxes display median (bold black line in middle of box) with upper and lower limits of IQR established by bounds of box. Each point is an individual crab righting time.

3.3. Hemolymph extraction and glucose assay

The experimental group had a higher average absorbance returned by the glucose assay. The range of absorbances of the experimental group was much larger than the range of absorbances of the control group (Figure 4). This meant that the control group was concentrated and consistent with a small spectrum of absorbances. The experimental group had a particularly high maximum (0.227 nm) and a low minimum (0.053 nm). No significant relationship or difference was detected.

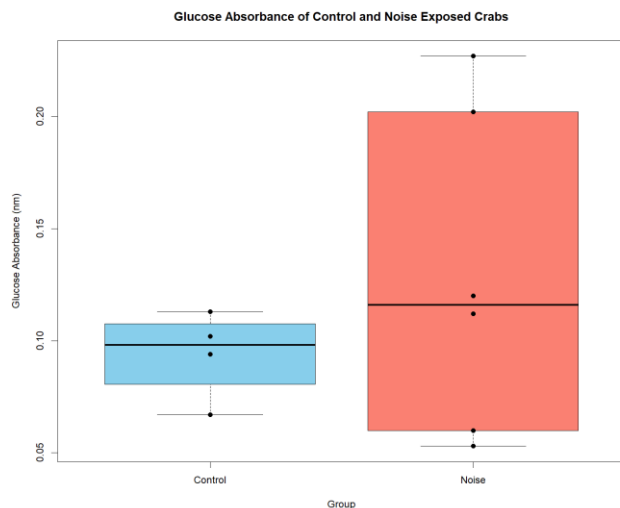


Figure 4: Absorbance of blood samples for control and experimental groups. Y axis displays glucose absorbance in nm, x axis shows which group is in the box above. Control in blue, experimental labeled as “noise” in red. Boxes display median (bold black line in middle of box) with upper and lower limits of IQR established by bounds of box. Each point is an individual crab righting time.

4. Discussion

Looking at the resazurin fluorescence is what helps us investigate whether respiration rate was elevated as a result of acute noise exposure. When an organism submerged within working resazurin solution respires, it removes oxygen from the solution. As oxygen is removed, it changes to resorufin, which is a pink color, when the original solution color is a deep blue (Sarker et al., 2007). Because of this, increased respiration causes the solution to become increasingly more pink over time, altering fluorescence. Fluorescence increased for both the experimental and control groups, which tells us that respiration was occurring, but the higher fluorescence for the control group is the opposite of

what was expected. The control group was respirating at a higher rate. There are a few possible explanations for this outcome. To begin with, there were 3 total molts that occurred in the control group, one of which occurred within the resazurin solution. 2 of these molts resulted in mortality. Molts could have been putting greater physiological stress on the crab body, requiring increased respiration as a result, especially when the molt happened mid-data collection. This idea is supported by the fact that 2 of the crabs that molted didn’t survive, meaning their fitness may have been lower. 0 molts occurred in the experimental group, and there were 0 mortalities in the experimental group, so fitness may not have been an issue. Alternatively, the music may simply lower the respiration rate of the crabs, a completely unexpected outcome. However, our results were not significant one way or the other, meaning we cannot draw any concrete conclusions from this data.

The righting time data was the one response parameter that suggested any sign of physiological stress placed on the experimental crabs, but this was only a slight sign and not a significant conclusion. With intense, acute exposure to the music, the experimental crabs may have been “shellshocked,” unable to maneuver themselves to the extent of their normal ability when flipped, as made evident by the quicker response times of the control group. This idea is supported by the excluded outliers, who may have been extreme cases of this effect, or may have taken longer to right for other external reasons.

From the glucose assay, we can look at absorbance and conclude how much glucose the crabs were using during their noise exposure. Higher absorbance indicates more glucose present within the blood, meaning less glucose was used. An indicator of physiological stress would be low absorbance, meaning low glucose was available and most of it had already been used due to elevated metabolic need. Although the control group had slightly lower average and median absorbance values, we cannot conclude much from this

evaluation technique. The experimental group had both oddly high glucose usage paired with glucose usage even lower than the control group. There are a few reasons this test might have produced such unreliable results. First, there were only 10 data points for this test, 4 from the control group and 6 from the experimental group. Such a small sample size may have led to statistical error. Secondly, this kit was for human use, applied to crab blood samples. The principle was the same, but this may have resulted in error. Finally, quite likely, there is simply no relationship between acute noise exposure and elevated glucose usage.

When all data is considered, there is a clear lack of a relationship between acute noise exposure and physiological stress identified from this study. On a broad scale, this is inconsistent with the idea that noise pollution negatively harms marine animals (Duarte et al., [2021](#)). It is possible that invertebrates aren't damaged in the same way, or that the noise exposure in this study was not to the level that the organisms victim to real anthropogenic noise influence experience. On a narrower scale, this is inconsistent with the findings of Birch, which identified ship noise as an inducer of increased respiration for European green crabs (Birch et al., [2025](#)). This could be because of a difference in noise

type, exposure duration, a possible difference in response between species.

In conclusion, acute anthropogenic noise exposure was found to have no effect on the hairy shore crabs. Similarly, exposures of similar nature and short duration likely will not induce physiological stress on Dungeness crabs. This is not in any way to say that Dungeness crabs are safe from anthropogenic noise pollution. The scope of this study limits conclusions that may be drawn drastically. A lack of study surrounding noise pollution for invertebrates still remains an issue, especially with the Dungeness crab fishery in such a critical and vulnerable state. Moving forward, longer durations of noise exposure may be found to induce stress over a longer period. A length of two hours may not be enough to induce any stress, especially with a week placed between each treatment. When exposed for days or weeks at a time, stress may begin to become more visible. Furthermore, music may be an ineffective type of noise, and ship or construction noises may be more common and worth looking into in an attempt to ensure safety for the crabs within the fishery. Waterfront concerts are not as common of a threat as these other anthropogenic noises. In combating European green crab invasion, no measure is too precautionary to assure the safety of those it threatens.